



## Lab 1

1 OF 1 QUESTIONS REMAINING

### Test Content

#### Question 1

100 Points

Use the data in the folder STWIN\_00012.

1- Do FFT on the signal for all three directions of the two accelerometer. the microphone is one dimensional. use hanning window

IIS2DH\_ACC

ISM330DHCX\_ACC

IMP34DT05\_MIC

2- from DeviceConfig.json, find the dynamic range of the accelerometers in g and the sampling rate of each accelerometer ( in the same folder)

3- Do not select any frequency range for FFT, what is the maximum frequency you see on the FFT representation of each sensor?

remember this is for the case where you increase the rpm from zero to max and back to zero

4-plot FFT of the signal

5-plot waterfall of each signal

please submit your work as a PDF

*Use the editor to format your answer*